

Experiment 44: Molecular models

Notes

Organic molecules that have the same molecular formula but different physical structures are called structural isomers

You can purchase petrol from a service station with different octane ratings such as 91, 95 and 98. The octane rating of petrol is its anti-knock rating or detonation resistance. 'Detonation resistance' is about burning more slowly; overly fast burning (detonation) creates a very fast expansion of gas, pushing unevenly on the piston and leading to lower efficiency and possible engine damage. In general, higher octane ratings give slower, smoother combustion. Modern high performance motors require the higher octane fuels. Octane rating is determined by comparing the fuel to a mixture of iso-octane (2,2,4-trimethylpentane, a **structural isomer** of octane) and normal heptane (the straight chain isomer of heptane). Iso-octane is given an octane rating of 100 and heptane is given an octane rating of zero. 91-octane petrol, for example, has the same anti-knock rating as a mixture of 91% (by volume) iso-octane and 9% (by volume) heptane. This does not mean, however, that the petrol contains these hydrocarbons in these proportions.

This experiment will help you to understand what isomers are and how 2,2,4-trimethylpentane and heptane are named and drawn. You will construct molecular models of, draw structures of and name some simple organic compounds, most of which are derived from natural gas and petroleum refining.

To draw three dimensional diagrams of models you should consider using the following method to represent bonds in molecules:

Bond Orientation	Symbol
In the plane of the page	—
Into the page	
Out of the page	↗

For example a three dimensional molecule like ammonia can be drawn as



Where one hydrogen is in the plane of the page, one is into the page and one is out of the page.

Equipment

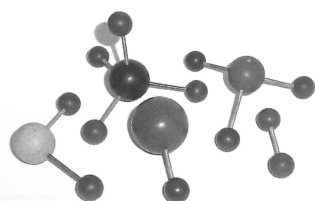
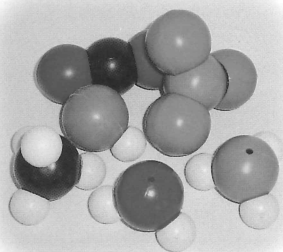
molecular model kit, or polystyrene balls and toothpicks

Procedure

A. alkanes

Methane [CH_4]

1. Use the molecular model kit provided to construct a model of the molecule.
2. Carefully observe the model of the molecule. Does it look the same from all sides? Describe the shape.
3. Draw a diagram of the model of the molecule.



Ethane [C₂H₆]

- Remove one hydrogen atom from your model of the methane molecule and replace it with another CH₃ group. This group should be identical to the one you hold. Are the bond angles still the same as for methane?
- Rotate the ends of the molecule about the C-C bond. Record your observations.
- Draw a diagram of the model.

Dichloroethane [C₂H₄Cl₂]

- Remove two hydrogen atoms from your model of the ethane molecule and replace them with two chlorine atoms.
- Construct another dichloroethane molecule different from the one you have already made. These two molecules are structural isomers. They have the same molecular formula but different structural formulas, that is, the order of attachment of atoms is different.
- Draw diagrams of the two models and label them 1,1-dichloroethane and 1,2-dichloroethane, as appropriate.

Hexane [C₆H₁₄] and Cyclohexane [C₆H₁₂]

- Construct a model of straight chain hexane.
- Write the structural formula for straight chain hexane.
- Write the structural formulas of four additional isomers with formula C₆H₁₄. Two of these isomers have five-carbon chains and two have four-carbon chains. Write the systematic name for these isomers.
- Arrange the model of straight chain hexane so that the two terminal carbon atoms are adjacent. Remove one hydrogen atom from each terminal carbon and join the two carbon atoms to form a cyclic structure. Is the molecule planar? Describe the shape.
- Draw a diagram of the structure.

Procedure**B. alkenes****Ethene [C₂H₄]**

- Construct a model of this molecule.
- Try to rotate the two ends of the molecule. What prevents rotation about the carbon-carbon double bond?
- Draw a diagram of your model.

Dichloroethene [C₂H₂Cl₂]

- Remove two hydrogen atoms from your model of the ethene molecule and replace them with chlorine atoms.
- Make another model of the ethene and by replacing two hydrogen atoms with two chlorine atoms, construct a different isomer of C₂H₂Cl₂.
- Construct a third isomer of C₂H₂Cl₂.
- Draw diagrams of the three isomers and write their names.

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Hexene [C_6H_{12}] and Cyclohexene [C_6H_{10}]

8. Draw diagrams of three straight chain isomers of hexene. Draw their structures and write their names.
9. Construct a model of one of these isomers of hexene.
10. Convert this to a model of cyclohexene in the same way in which you constructed a model of cyclohexane.
11. Draw a diagram of the model.

Procedure

C. Alkynes (optional extra)

Ethyne [C_2H_2]

1. Construct a model of this molecule.
2. Draw a diagram of the model.

Processing of results and questions

1. Using examples from this activity distinguish between structural isomerism and geometric isomerism.
2. Which groups of aliphatic (straight chain) hydrocarbons display geometric isomerism?
3. Explain why some groups of aliphatic (straight chain) hydrocarbons do not display geometric isomerism.
4. Research differences in the physical properties between two structural isomers and two geometric isomers. Write an account of your findings.